

Mid-Evaluation Guide: Does Biodiversity Track Ecosystem Productivity?

Course Project 9 · mid evaluation: project, approach, literature, impact, expected results · 2026-10-07

This guide covers exactly what the mid evaluation asks: what the project is, how we are approaching it, the literature and sources we use, the impact, and the results we expect. The short "early indications" box in section 7 is only for if an examiner asks how far the work has got.

1. The project in one minute (open with this)

"We are testing whether places with more bird species are also more productive, meaning their plants fix more carbon, across South and Southeast Asia. We combine 71.7 million citizen-science bird records from GBIF with NASA satellite productivity data on a grid of 884 one-degree cells. The catch is that citizen-science records pile up where birdwatchers go. So the core of our approach is correcting for that sampling bias before asking the question, and showing how much the answer changes once we do."

2. What is the project?

2.1 The question

Does biodiversity track ecosystem productivity at landscape scale, once the uneven sampling of biodiversity data is corrected?

2.2 Why ask it now

- **Policy.** In December 2022 the 196 Parties to the Convention on Biological Diversity adopted the **Kunming–Montreal Global Biodiversity Framework** (Decision 15/4). Its Target 19 commits to mobilising **at least USD 200 billion a year by 2030** for biodiversity. Part of the case for that money is that biodiverse ecosystems fix and store more carbon.
- **Economics.** The World Economic Forum estimates that **over half of world GDP (about \$44 trillion)** depends moderately or highly on nature (WEF, 2020).
- **A scale gap in the evidence.** The best evidence that biodiversity raises productivity comes from experiments and forest plots. Conservation, however, is planned over landscapes and regions, so the question needs testing at that scale.
- **A data problem.** Citizen-science data (GBIF, eBird) are increasingly used to monitor biodiversity targets, but they are heavily biased towards accessible, well-surveyed places. How much that bias distorts conclusions is itself an open question.

2.3 Hypotheses

- **H1:** effort-corrected biodiversity is positively associated with productivity (gross primary productivity, GPP) after controlling for climate and restricting to natural land cover.
- **H0:** no meaningful association remains once sampling effort, climate and land cover are accounted for.

Framing note: we test an **association**. Whether biodiversity *causes* higher productivity, or productivity supports more species, is a question our design can inform but not settle (section 7).

2.4 Three extensions

	Question	Why it matters
A Naive vs corrected	How different is the answer with raw species counts versus effort-corrected richness?	Shows directly how much citizen-science bias changes conclusions
B Linear vs hump	Does productivity rise steadily with diversity, or peak and fall?	A live debate in the literature (section 4)
C Three case studies	Does the pattern hold in contrasting landscapes: Western Ghats (wet forest hotspot), Thar Desert (arid), Eastern Himalaya (high diversity, under-sampled)?	Tests whether a regional pattern holds on the ground

2.5 Scope

Item	Detail
Region	5–31°N, 68–102°E: South and Southeast Asia (India, southern China, Myanmar, Thailand, Pakistan, Nepal, Bangladesh, Sri Lanka and others)
Unit	1° × 1° grid cells (~111 km), 884 cells, 630 on land
Biodiversity	Birds (class Aves), the most consistently recorded group
Period	Bird records 2000–2024; satellite and climate averages 2019–2023

3. How we are approaching it

3.1 The pipeline

1 · Grid 884 one-degree cells over the study region	2 · Bird data GBIF records per cell: total records and the species list with counts	3 · Bias correction Rarefaction: expected species in an equal sample of 250 records; effort kept as a covariate
4 · Satellite & climate MODIS productivity, ERA5-Land temperature and rainfall, ESA land cover, via Google Earth Engine	5 · Mask & merge Drop cropland, built-up and water cells; join everything per cell	6 · Models Naive vs corrected regression (A); linear vs quadratic (B)
7 · Spatial checks Moran's I, spatial models, clustering by ecoregion, other grid sizes	8 · Case studies Western Ghats, Thar Desert, Eastern Himalaya (C)	9 · Write-up Report, policy implications, presentation

3.2 Data sources

Data	Provider	Resolution	Role
Bird occurrences	GBIF (98.8% of records come from eBird)	point records	Species richness and survey effort
GPP and NPP	NASA MODIS MOD17A3HGF v061	463 m, yearly	Productivity (the response variable)

Data	Provider	Resolution	Role
NDVI and EVI	NASA MODIS MOD13Q1 v061	232 m, 16-day	Alternative productivity measures
Temperature, rainfall	ECMWF ERA5-Land	~11 km, monthly	Climate controls
Land cover	ESA WorldCover v200 (2021)	10 m	Removing cropland, cities and water
Elevation	NASA SRTM	30 m	Topography as a control
Ecoregions	RESOLVE Ecoregions 2017	polygons	Biogeographic units for clustering

3.3 Methods, and why each one

Method	What it does	Why we use it	Source
Rarefaction	Expected species count in a random sample of 250 records	Compares cells at equal survey effort, so heavily birded cells do not look richer just because they are watched more	Hurlbert (1971)
Effort covariate	Log of record count included in the regression	Current practice combines rarefaction with explicit effort	El-Gabbas (2026); Hughes et al. (2021)
Coverage standardisation (robustness)	Compares cells at equal sample completeness	Size-based rarefaction can distort differences between rich and poor communities	Chao & Jost (2012)
Land-cover mask	Keeps only natural-cover cells	Farm productivity reflects irrigation and fertiliser, not ecology	Plan, Step 4
Climate controls	Temperature and rainfall in the model	Climate drives both diversity and productivity	Grace et al. (2016)
Regression with robust errors	OLS with heteroskedasticity-robust standard errors	Ecological data rarely have constant variance	Plan, Step 6
Linear vs quadratic (AIC)	Compares a straight line with a curve	Tests the hump-shaped hypothesis directly	Fraser et al. (2015); Liang et al. (2016)
Spatial autocorrelation checks	Moran's I; spatial lag and spatial error models	Neighbouring cells are not independent; ignoring this inflates significance	Ploton et al. (2020); Kissling & Carl (2008)
Clustering by ecoregion	Standard errors that allow correlation within biogeographic units	The plan asks for clustering "by biogeographic sub-region"	Dinerstein et al. (2017)
Spatial block cross-validation	Tests prediction on whole held-out regions	Random cross-validation is over-optimistic for spatial data	Roberts et al. (2017)
Grid-size robustness	Repeats at 0.5° and 2°	Results should not hinge on one grid	Plan, Step 8; Gonzalez et al. (2020)

3.4 Tools

Tool	Used for	Why
Python (pandas, numpy)	All data handling	Standard and readable by the whole team

Tool	Used for	Why
GBIF occurrence API	Bird records per cell	Free, global, no login. One request per cell returns the full species list with counts.
Google Earth Engine	Satellite, climate and land-cover values per cell	Processes terabytes of imagery on Google's servers; free for academic use
statsmodels	Regression, robust and clustered errors	Standard statistics library
PySAL (libpysal, esda, spreg)	Neighbour matrices, Moran's I, spatial regression	The standard Python spatial statistics toolkit
matplotlib	Maps and figures	Print-quality static figures

3.5 Decisions made so far, and why

- **Birds, not plants:** a live check across the three case-study regions found about **200× more records per cell** for birds (e.g. Deccan median 29,642 vs 145). Plant richness could not be corrected reliably in most cells.
- **Thar Desert replaces the Deccan Plateau:** all 25 Deccan land cells are dominated by cropland (77% converted), so none remain after removing farmland. The Thar was chosen by a systematic scan as the driest region that is still largely natural. The Deccan's conversion is itself a finding about land use.
- **Water also removed:** the plan removed cropland and cities. Sea and lake cells were added to the mask, because they would otherwise enter as "natural" cells with near-zero productivity and create a false signal.
- **Ecoregions as the biogeographic units** for the plan's Step 8 clustering.

4. The literature we have used

4.1 How the literature is organised, and what it means for us

Theme	Key studies	What they show	What it means for our design
From experiments to real ecosystems	Liang et al. 2016; Duffy et al. 2017; Chen et al. 2023; Deng et al. 2025	Biodiversity is positively linked to productivity and carbon in real forests, not just experiments; canopy complexity is one mechanism	Our H1 expects a positive association
The shape is disputed	Adler et al. 2011; Fraser et al. 2015; Grace et al. 2016; Gonzalez et al. 2020	Null, linear and hump-shaped results all exist; shared environmental drivers and spatial scale explain much of the disagreement	Extension B; climate controls; grid-size checks
What "productivity" means	Running et al. 2004; Liang et al. 2016; Chen et al. 2023; Kothandaraman et al. 2020	Studies mix fluxes (GPP), growth rates and carbon stocks, which need not move together	We use a flux (GPP) and say so; NPP as a check
Satellite productivity at landscape scale	Running et al. 2004; Oehri et al. 2017, 2020	Satellite productivity works as a biodiversity-study response, so far only with evenly sampled national inventories	Our design uses it with <i>unevenly</i> sampled data, which is the novelty
Sampling bias in occurrence data	Beck et al. 2014; Engemann et al. 2015; Hughes et al. 2021; Hurlbert 1971; El-Gabbas 2026; Ploton et al. 2020	GBIF is strongly biased; raw richness tracks sampling intensity; corrections exist; spatial validation matters	Extension A; rarefaction plus effort covariate; spatial checks

Theme	Key studies	What they show	What it means for our design
Regional evidence (Western Ghats)	Kothandaraman et al. 2020; Najeeb et al. 2025; Ramachandra & Bharath 2020	Diversity is linked to carbon storage, but stand structure matters more; all plot-based	Extension C; a grid-based regional test

4.2 The gap we address

No study for this region combines **effort-corrected, citizen-science biodiversity** with **satellite productivity** at grid-cell resolution, restricted to natural land cover, while reporting **spatial diagnostics**. (The literature review states this for South Asia; our grid also covers Southeast Asia.)

4.3 Theory and methods literature used in the approach

- **Spatial statistics:** Hawkins et al. (2007), Kissling & Carl (2008), Bini et al. (2009), Halleck Vega & Elhorst (2015), Roberts et al. (2017). They tell us how to model spatial dependence and how to validate predictions honestly.
- **Scale and species pools:** Chase & Leibold (2002), Ricklefs (1987), Cornell & Harrison (2014). The same relationship can look different between and within regions, because regional species pools differ.
- **Species–energy theory:** Evans et al. (2005), Hurlbert (2004), Coops et al. (2009). The leading *alternative* explanation: productive places support more bird species, rather than the reverse.
- **Habitat heterogeneity:** Stein et al. (2014). Topographic variety raises richness, so it is a control.
- **Coverage-based rarefaction:** Chao & Jost (2012).

Every reference in this guide was checked against Crossref for existence and details, and its abstract was read.

5. Sources and full citations

5.1 Literature review (the 20 submitted references)

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12. Hurlbert, S. H. (1971). The nonconcept of species diversity: A critique and alternative parameters. *Ecology*, 52(4), 577–586. <https://doi.org/10.2307/1934145>
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5.2 Theory and methods references

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5.3 Data sources

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5.5 Source notes (know these before you are asked)

- **GBIF requires a citable DOI.** Our data came from live API queries, which do not create one. A DOI will be registered (a GBIF download or "derived dataset") before the final report.
- **Corrected from the plan:** the plan attributes "~55% of global GDP" to IUCN. The figure is the World Economic Forum's (2020): over half of world GDP, about \$44 trillion.
- **Not used unless verified:** the plan also mentions a Carbon Brief article (journalism, not peer-reviewed) and an unnamed "2025 subtropical Chinese forest study". We have not identified or read the latter, so neither is cited.

6. Impact

Level	Impact	Honest limit
Scientific	Tests a biodiversity–productivity claim at landscape scale, with open data, in an under-studied region containing two global biodiversity hotspots (Western Ghats, Eastern Himalaya)	Correlational: it can test whether the pattern exists and at what scale, not prove cause
Methodological	A reproducible open pipeline (GBIF + Google Earth Engine) that measures how much citizen-science bias changes the answer, with a fast way to get bias-correctable data per cell	Shown for birds, the best-recorded group; other taxa are likely worse
Policy	Evidence on whether the "biodiversity helps carbon uptake" argument behind biodiversity finance holds at the scale where money is spent	We measure carbon <i>uptake</i> (a flux), not carbon storage
Monitoring	The GBF tracks progress partly with citizen-science data. Showing the size of the bias, and correcting it, matters for how that monitoring is read	—
Land use	Shows how much natural cover remains for such comparisons, e.g. the Deccan Plateau is almost entirely converted to farmland	A 1° grid is coarse for local planning

7. Expected results

Each expectation comes from the literature in section 4. The plan's risk table treats a weak or null result as a valid finding, not a failure.

#	We expect	Because	What would surprise us
1	Raw species counts mostly reflect survey effort	GBIF bias is strong and systematic (Beck 2014; Engemann 2015; Hughes 2021)	Raw and corrected richness telling the same story
2	A positive association after correction, but weaker than the naive one	Positive in real forests (Liang 2016; Duffy 2017; Oehri 2017), but climate drives both variables (Grace 2016)	No association at all, or a negative one
3	Climate explains much of the raw pattern	Shared environmental drivers (Grace 2016)	Richness mattering more than climate
4	A straight or levelling-off line, not a hump	Forest studies find positive, decelerating curves (Liang 2016; Deng 2025); the hump comes mainly from grasslands (Fraser 2015)	A clear hump

#	We expect	Because	What would surprise us
5	Strong spatial autocorrelation	Large-scale ecological maps nearly always show it (Ploton 2020; Hawkins 2007)	Independent residuals
6	A stronger pattern between regions than within them	Regional species pools and scale dependence (Ricklefs 1987; Chase & Leibold 2002; Gonzalez 2020)	The same strength within and between regions
7	Case studies: Western Ghats productive and species-rich; Thar low on both; the Eastern Himalaya showing the biggest naive-vs-corrected gap (under-sampled)	The plan's predictions; Western Ghats plot evidence (Kothandaraman 2020)	The ranking unchanged by correction
8	Direction will stay open	Birds are consumers; species–energy theory predicts productivity → richness (Evans 2005; Hurlbert 2004; Coops 2009)	—

What each outcome would mean

- *Clear positive after correction* → supports biodiversity-based conservation arguments at landscape scale (as an association).
- *Weak, or mostly between regions* → the policy argument holds at best regionally, and is driven by species pools and climate.
- *Null* → valid and publishable. It would match the scale-dependence literature and caution against blanket "biodiversity = carbon" claims.

Early indications — only if asked how far you have got. The data are assembled and the first analyses are run. They are consistent with expectations 1, 4, 5, 6 and 7. Raw richness correlates 0.81 with survey effort (0.24 after correction). The corrected association is positive but much weaker within ecoregions than between them. There is no hump. Spatial autocorrelation is strong (Moran's I 0.72). The case-study ranking changes after correction. These are preliminary and will be reported in full in the final report.

8. Progress and next steps

- **Done:** grid; bird data (71.7 million records); bias correction; satellite, climate and land-cover extraction; merged dataset; first models; spatial checks; case studies; figures; literature review (submitted).
- **Next (weeks 7–8):** the written report and policy section; the presentation; registering the GBIF DOI; adding the policy sources (5.4) to the introduction; agreeing as a team how to frame H1 (association vs cause).

9. Questions you may be asked

What is your project, in one sentence? Whether places with more bird species are also more productive across South and Southeast Asia, once citizen-science sampling bias is corrected.

What is new about it? Earlier landscape studies used evenly sampled national inventories (Oehri 2017). We use globally available but biased citizen-science data, correct the bias, and measure how much that correction changes the answer, in a region where this has not been done.

Why birds? They are the best-recorded group: about 200× more records per cell than plants in our check. 98.8% of our records come from eBird.

What is the main methodological challenge? Sampling bias: records concentrate where birdwatchers go. We use rarefaction (compare cells at an equal sample of 250 records) plus survey effort as a covariate.

Why satellite GPP? It is the standard global, consistent measure of how much carbon plants fix each year (Running et al. 2004). It is a flux, not a carbon stock, and we are explicit about that.

Why a 1° grid? Large enough that each cell has many records to correct for bias; small enough to give hundreds of cells. We will check 0.5° and 2° as well.

Why remove cropland? Farm productivity reflects irrigation and fertiliser, not natural ecology. We also remove water.

Why the Thar instead of the Deccan? Every Deccan cell is mostly farmland, so none survive the natural-land filter. The Thar is the driest region that is still mostly natural.

Which literature most shaped your design? Liang et al. (2016) and Duffy et al. (2017) for the expected positive link; Hughes et al. (2021) and Engemann et al. (2015) for the bias problem; Fraser et al. (2015) vs Liang et al. (2016) for Extension B; Ploton et al. (2020) for spatial checks.

Is the relationship causal? We test association. Direction is open: species–energy theory suggests productivity may support more bird species, rather than the reverse.

What if you find no relationship? That is a valid result. The plan says so explicitly, and it would match the scale-dependence literature (Gonzalez et al. 2020; Chase & Leibold 2002).

What is the impact? It tests, at the scale where conservation money is spent, an argument used to justify biodiversity finance. It also shows how much uncorrected citizen-science data can mislead biodiversity monitoring.

Are your data citable? All satellite and climate products have DOIs (section 5.3). GBIF requires a DOI for our bird data, which we will register before the final report.

10. Short glossary

- **Species richness:** number of species.
- **Rarefaction:** expected species count at a fixed sample size (here, 250 records).
- **Survey effort:** number of records in a cell.
- **GPP / NPP:** carbon fixed by plants per year, before / after plant respiration.
- **Land-cover mask:** removing cells dominated by cropland, cities or water.
- **Spatial autocorrelation / Moran's I:** nearby cells being similar; Moran's I measures it (0 = none).
- **Ecoregion:** a biogeographic unit with a distinctive community of species.
- **AIC:** model-fit score penalising complexity; lower is better.
- **Association vs causation:** two things varying together vs one causing the other.

Appendix — figures

The study area and what the two main data layers look like. Each has a CSV of its plotted values in [figures/](#).

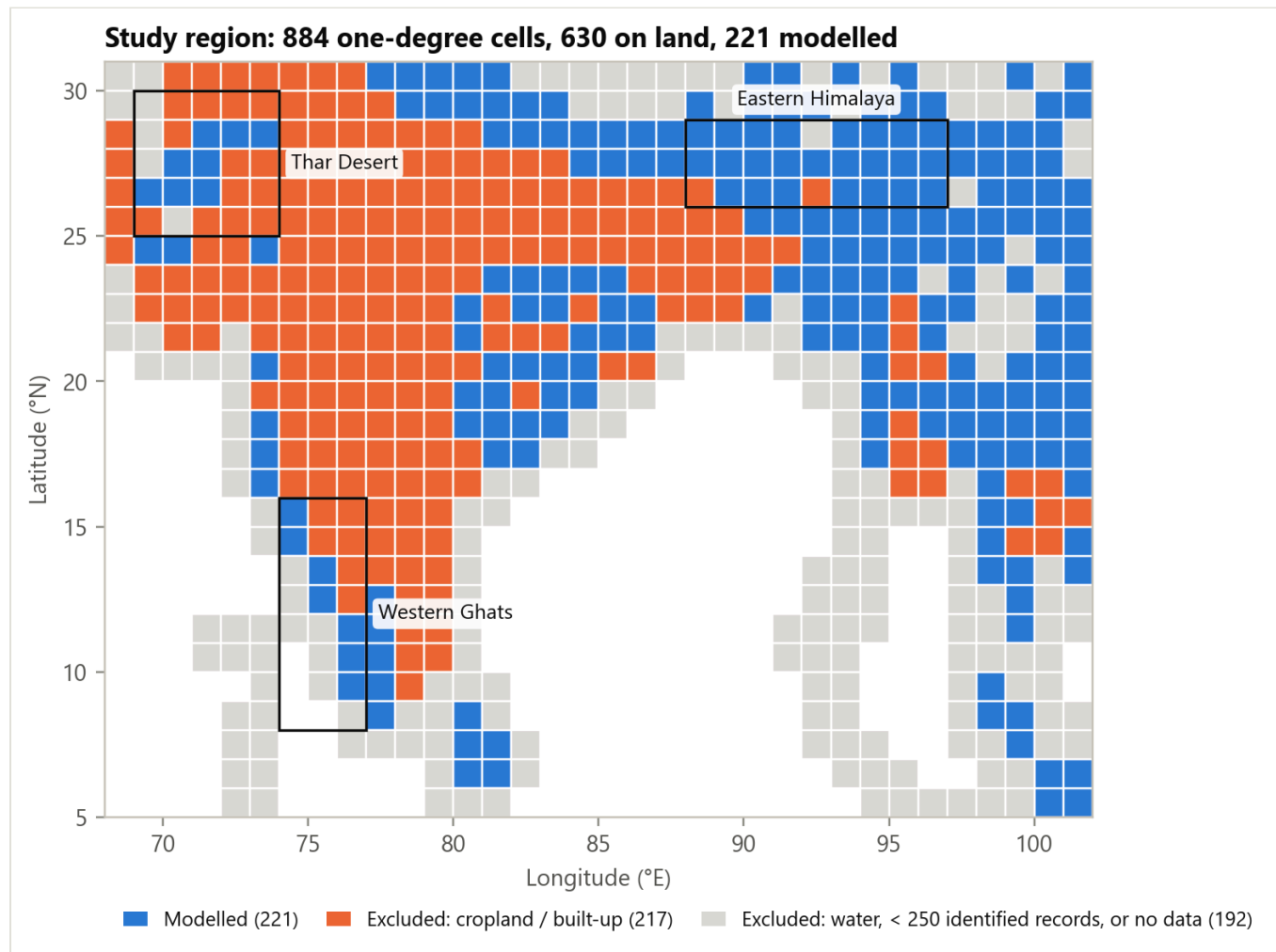


Fig 1 — Study region and mask. 884 one-degree cells, 630 on land, 221 modelled. Orange cells are cropland or built-up dominated and excluded; the Deccan disappears entirely. Boxes: the three case studies.

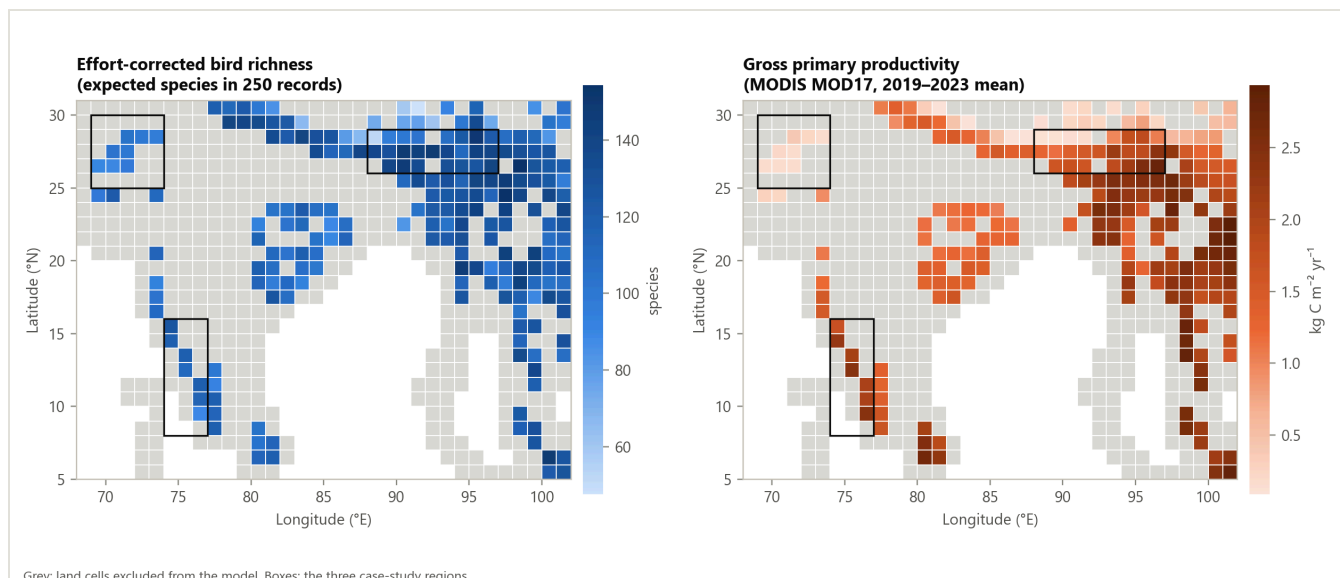


Fig 2 — Effort-corrected richness and GPP. Both are highest in the wet north-east (Eastern Himalaya, Myanmar). Grey: land cells excluded from the model.